Focused Hypothesis Report — *S. pombe* Mus81 mitochondrial localization & mtDNA function

Gene: mus81 (UniProt **P87231**, MUS81_SCHPO), *Schizosaccharomyces pombe* (NCBITaxon:284812) **Source review:** genes/SCHPO/mus81/mus81-ai-review.yaml — free-text hypothesis **Seed hypothesis:** *Mus81 has a genuine mitochondrial localization and a direct role in mitochondrial DNA metabolism, rather than the mitochondrial HDA signal being a high-throughput localization artifact.*

Executive Judgment

Verdict: Refuted (the mitochondrial/mtDNA claim is over-annotated; the HDA mitochondrion signal is best treated as a high-throughput localization artifact).

Four independent lines of evidence converge against the seed hypothesis:

1. **Annotation provenance.** QuickGO shows the mitochondrion term (GO:0005739, HDA) **and** the nucleus term (GO:0005634, HDA) for Mus81 both trace to a **single reference,** P 16823372 (Matsuyama et al. 2006 — genome-wide C-terminal YFP tagging of ~90% of the *S. pombe* proteome). This is one high-throughput imaging dataset that reported a dual/ambiguous signal. In contrast, every **low-throughput direct assay (IDA)** for Mus81 comes from dedicated papers and places it in **nuclear** complexes (Holliday junction resolvase complex GO:0048476, P 11719193 / 14527419; nuclear replication fork GO:0043596, P 17363897), and the curated UniProt subcellular-location statement is **Nucleus** only.

2. **Sequence (with positive control).** The Mus81 N-terminus has **no canonical mitochondrial targeting presequence**, shown directly by side-by-side comparison with the bona fide mitochondrial resolvase Cce1/Ydc2 (UniProt Q10423, curated *Mitochondrion*):

Protein (first 30 aa)	Arg	R+K	D+E	Net charge	1st acidic pos	GRAVY
Cce1/Ydc2 (Q10423, genuine mito) `MATVKLSFLQHICKLTGLSRSGRKDELLRR`	4	7	2	**+5**	**25**	−0.25
Mus81 (P87231, claimed mito) `MDCGNPLFLQWIQEWMEESTRRFPKSYQTW`	2	3	4	**−1**	**2**	−0.92

Cce1/Ydc2 has the textbook Arg-rich, net-positive presequence with no acidic residue until position 25; Mus81 is net-**negative** with an acidic residue (Asp) immediately at position 2 — the opposite of an import signal.

1. **Biological redundancy.** *S. pombe* already possesses a **dedicated mitochondrial Holliday-junction resolvase, Ydc2/SpCce1** (RuvC/CCE1 family), which localizes exclusively to mitochondria and is the enzyme responsible for mtDNA maintenance (loss → mtDNA aggregation/depletion). Mus81 is a structurally unrelated **nuclear XPF-family** endonuclease. There is no functional niche that requires Mus81 in mitochondria, and no primary literature reports any mitochondrial or mtDNA role for Mus81.

2. **No conservation.** Direct QuickGO cross-check: *S. cerevisiae* MUS81 (Q04149) and human MUS81 (Q96NY9) carry **only nuclear** cellular-component annotations and **no** mitochondrion term. The mitochondrion call is unique to *S. pombe* Mus81 and to one HDA reference — not a conserved feature of the MUS81 family.

Most important caveat: This assessment is negative/inferential. An HDA call is weak positive evidence, not proof of absence; a genuine minor mitochondrial pool cannot be excluded without a targeted, endogenously-tagged localization experiment. The sequence check is a heuristic (TargetP/MitoFates could not be executed programmatically here). But the balance of evidence is strongly against a genuine, functional mitochondrial role.

Evidence Matrix

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
UniProt **P87231** (database)	Review/ database	Qualifies	Where is Mus81 annotated?	Curated location = **Nucleus** only; mitochondrion = **HDA**; nucleus complexes = **IDA**	*S. pombe*, curated record	High for provenance; HDA is low-specificity
QuickGO / P 16823372 (Matsuyama 2006)	Localization (high-throughput)	Refutes (artifact source)	Origin of the mito call	Mitochondrion **and** nucleus HDA calls both come from the **same** genome-wide C-terminal YFP study; a single ambiguous HT dataset	*S. pombe* proteome screen	High for provenance; HT dual-localization is low-specificity
This report (computed, N-term MTS + positive control)	Computational	Refutes	Does Mus81 carry an MTS?	Mus81 first-30: net **−1**, acidic at pos 2, GRAVY −0.92 → **no MTS**; positive control Cce1/ Ydc2 (Q10423): net +**5**, first acidic pos 25 → canonical MTS	Sequence heuristic + control	Medium-high; heuristic, but validated against a true mito protein
PMID **11719193** (Boddy 2001)	Direct assay / interaction	Refutes (context)	Mus81 identity/ locale	Mus81-Eme1 identified as **nuclear** structure-specific endonuclease (basis of IDA GO:0048476)	*S. pombe*	High

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
23584455	Direct assay / mutant	Refutes (context)	Mus81 core function	Mus81-Eme1 is a **nuclear** HJ resolvase activated by CDK/ATR (Eme1 phospho) to prevent chromosomal rearrangements	*S. pombe*	High
22855558	Mutant phenotype	Refutes (context)	Mus81 substrate/ locale	Resolves **meiotic nuclear** joint molecules (crossovers); regulated by Smc5/6	*S. pombe* meiosis	High
19470480	Mutant phenotype	Refutes (context)	Mus81 role	Mus81-Eme1 generates **nuclear** meiotic crossovers at DSBs	*S. pombe* meiosis	High
23982516	Structural / biochem	Refutes (context)	Mus81 mechanism	WH domain of MUS81 binds branched DNA; yeast phenocopy links to nuclear DNA-damage sensitivity	Human + *S. pombe*	High
10954073	Localization + mutant	Competing	Which enzyme handles	**SpCCE1/Ydc2-GFP localizes exclusively to mitochondria**; ydc2Δ →	*S. pombe*	High — identifies the true mt resolvase

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
			mtDNA junctions?	mtDNA aggregation		
12823554	Mutant / rescue	Competing	Ydc2 mtDNA role	ydc2Δ → **mtDNA depletion**, rescued only by active full-length Ydc2; mutants localize to mitochondria	*S. pombe*	High
11726496	Structural	Competing	Nature of mt resolvase	Ydc2 = eukaryotic RuvC-family mt HJ resolvase (crystal structure)	*S. pombe*	High
9421521 / 9343409	Biochemical	Competing	Ydc2 identity	Ydc2 is the functional *S. pombe* CCE1 homolog, a mt HJ resolvase	*S. pombe*	High
QuickGO orthologs **Q04149** (*S. cerevisiae*), **Q96NY9** (human)	Structural/ evolutionary (comparative)	Refutes	Is the mito call conserved?	Both orthologs have **exclusively nuclear** CC terms; **no** GO:0005739 in either — mito call is *S. pombe*-only & single-study	Budding yeast + human	High for provenance; annotation completeness caveat
PubMed "Mus81	Absence of evidence	Refutes		**No papers returned**	—	Absence, not proof

Citation (PMID)	Evidence type	Stance	Claim tested	Key finding	Context	Confidence / limitations
mitochondria/ mtDNA"			Any mt role for Mus81?			

GO Curation Implications (leads — require curator verification)

- **GO:0005739 `mitochondrion` (CC), evidence HDA:PomBase — lead: consider REMOVE or DO-NOT-PROPAGATE / NOT-core.** It is unsupported by any focused assay, contradicted by the curated Nucleus location, by the absence of an MTS, and by the existence of a dedicated mitochondrial resolvase (Ydc2). At minimum it should not be treated as representing a biological function. If retained, flag as high-throughput-only and non-core.
- **Retain nuclear CC terms:** GO:0048476 `Holliday junction resolvase complex` (IDA) and GO:0043596 `nuclear replication fork` (IDA); GO:0005634 `nucleus` is well supported. These are the genuine location.
- **MF:** retain `crossover junction endonuclease activity` / structure-specific (5'-flap / branched DNA) endonuclease activity.
- **BP:** retain nuclear DNA repair / recombination / replication-fork processing / meiotic recombination. Do **not** add mtDNA-metabolism BP terms to Mus81.
- Avoid a bare "protein binding" fallback — the informative, supported terms are the nuclear resolvase MF/CC/BP set above.

Mechanistic Scope

Direct molecular activity (supported): Mus81, in a heterodimer with Eme1, is a Mg^{2+}-dependent structure-specific endonuclease that incises branched DNA (3'-flaps, D-loops, replication forks, nicked Holliday junctions) in the **nucleus**, resolving recombination/ replication intermediates. Activity is cell-cycle/DNA-damage regulated via CDK/ATR phosphorylation of Eme1.

Claimed but unsupported (seed hypothesis): import into mitochondria and direct action on mtDNA. This is not an observed direct activity; it is inferred only from a high-throughput localization dataset. The corresponding mtDNA-junction-processing function in *S. pombe* is performed by the unrelated Ydc2/SpCce1.

Conflicts and Alternatives

- **Dedicated-enzyme alternative (strongest):** Ydc2/SpCce1 already fulfills mitochondrial HJ resolution/mtDNA maintenance; a second, unrelated nuclease for the same niche is biologically unnecessary and unreported.
- **Artifact of high-throughput tagging:** N-terminal or over-expression YFP/GFP tagging commonly yields low-level spurious mitochondrial signals; the HDA code is exactly this class of data.
- **Paralog/ortholog carry-over (checked directly):** QuickGO CC annotations for *S. cerevisiae* MUS81 (Q04149) and human MUS81 (Q96NY9) are **exclusively nuclear** (nucleus, nuclear chromosome, nuclear replication fork, HJ resolvase complex, telomeric/chromosomal region, nucleoplasm) with **no mitochondrion (GO:0005739)** term in either. The mitochondrion call exists **only** for *S. pombe* Mus81 and **only** via the single HDA reference

  P 16823372

  — i.e., it is species-specific and not conserved, arguing against genuine, conserved mitochondrial biology.
- **Not excluded:** a small, functionally minor mitochondrial pool, or a stress-specific relocalization, cannot be ruled out by these data alone.

Knowledge Gaps

1. **Provenance of the exact HDA dataset — RESOLVED.** Checked: QuickGO → the mitochondrion HDA (and nucleus HDA) call both derive from

   P 16823372

   (Matsuyama et al. 2006), a **C-terminal (3′) YFP** genome-wide screen. Residual gap: raw image for Mus81 in that dataset not inspected here; a C-terminal tag would not mask an N-terminal MTS, so the mito signal most plausibly reflects HT imaging ambiguity/overexpression. Resolve: inspect the original localization image/record.

2. **Endogenous localization.** Checked: curated Nucleus statement + IDA nuclear complexes. Gap: no endogenously-tagged, mitochondria-co-stained Mus81 image was located. Resolve: mNeonGreen knock-in Mus81 + MitoTracker/mtDNA co-staining and fractionation.
3. **Formal MTS prediction.** Checked: manual N-terminal charge/hydropathy heuristic (no MTS). Gap: TargetP2/MitoFates/DeepMito not runnable programmatically here. Resolve: run these predictors on P87231.
4. **mtDNA phenotype of mus81Δ.** Checked: literature — none reported. Gap: no direct test of mtDNA copy number/integrity in mus81Δ. Resolve: qPCR mtDNA copy number and mtDNA topology in mus81Δ vs ydc2Δ.

Discriminating Tests

1. **Endogenous knock-in localization** + **fractionation:** tag chromosomal *mus81* with a fluorophore; co-image with a mitochondrial marker and quantify Mn-fold enrichment; biochemically fractionate and immunoblot for Mus81 in purified mitochondria (with nuclear/cytosolic controls).
2. **mtDNA functional assay:** measure mtDNA copy number, aggregation, and petite/respiration phenotypes in *mus81Δ* vs *ydc2Δ* vs WT. Ydc2Δ shows mtDNA aggregation/depletion; a genuine Mus81 role predicts a (possibly redundant) mtDNA phenotype, whereas an artifact predicts none.
3. **MTS predictors** (TargetP2, MitoFates, DeepMito) on P87231 and a fused N-terminal-Mus81(1–40)-GFP import assay.
4. **Epistasis:** *mus81Δ ydc2Δ* double mutant — synthetic mtDNA defect would argue for a real (backup) mitochondrial role; no aggravation argues against.

Curation Leads (require curator verification)

- **Action change:** Mark GO:0005739 `mitochondrion` (HDA) as **non-core / candidate removal** for Mus81; keep nuclear CC/MF/BP terms as the core annotation.
- **Candidate supporting references (verify snippets):**

- PMID **10954073**: "*a SpCCE1-GFP fusion localises exclusively to the mitochondria of S. pombe*" — identifies the true mitochondrial resolvase (competes with the Mus81 mt claim).
- PMID **12823554**: "*Cells that lacked Ydc2 showed a significant depletion of mtDNA content*" — the mtDNA-maintenance function is Ydc2's, not Mus81's.
- PMID **23584455**: DNA-damage-induced activation of the **nuclear** Mus81-Eme1 resolvase — Mus81's genuine role/location.
- UniProt **P87231**: curated SUBCELLULAR LOCATION = Nucleus; mitochondrion term is HDA only.
- **Suggested curator questions:** Which high-throughput study underlies the HDA mitochondrion call, and what tag/expression was used? Is there any endogenous, non-HT evidence for Mus81 in mitochondria?
- **Suggested experiments:** endogenous-tag co-localization + fractionation; mtDNA copy-number/topology in mus81Δ; MTS predictions and N-terminal import assay.

Provenance note

Computational analyses run here (code + raw outputs in the iteration transcript): (1) live UniProt REST fetch of P87231 (length 608; curated Nucleus; GO-CC evidence codes as tabulated); (2) **QuickGO annotation query** for P87231 → both mitochondrion and nucleus HDA terms map to P 16823372; nuclear IDA terms map to P 11719193/14527419/17363897; (3) N-terminal targeting-signal feature calculation for Mus81 vs the mitochondrial positive control Cce1/Ydc2 (Q10423): Mus81 first-30 net −1, acidic at pos 2, GRAVY −0.92; Cce1/Ydc2 net +5, first acidic pos 25, GRAVY −0.25. Writing to a provenance directory was blocked by sandbox permissions; results are reproducible from the recorded code. No result was fabricated; dedicated MTS predictor tools (TargetP2/MitoFates/DeepMito) could not be executed in this environment and are listed as gaps.
